

PEIYU CHEN

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EDUCATION

The University of Chicago

M.S. in Computational Analysis and Public Policy, GPA: 3.81/4.00

Chicago, IL

Expected June 2026

- Highlight Course: Human-Computer Interaction, Algorithm, Generative AI, Database, Machine Learning

Beijing Normal University

B.Econ. in International Economics and Trade, Minor: B.A. in Spanish, GPA: 85.4/100

Beijing, China

June 2024

Advisor: Prof. Lan-cui Liu

RESEARCH INTEREST

Human-AI Interaction; AI-mediated Social Interaction; Decision-making

RESEARCH EXPERIENCE

Data Science Institute, The University of Chicago

Chicago, IL

Data Science Clinic Fellow, *AICE: AI for Climate*

January 2026 - Present

- Built MonsoonBench, an open-source Python benchmark for monsoon onset prediction, integrating heterogeneous rainfall observations, forecast outputs, and standardized skill metrics
- Engineered reproducible evaluation pipelines enabling cross-model, cross-region, and cross-time comparisons

Oxford Computational Political Science Group (OCPSG)

Oxford, UK (Remote)

Research Associate, affiliated with DPIR, University of Oxford

December 2025 - Present

- Designed experimental protocols and cross-pollination treatments for AI-mediated deliberation on Harmonica, with a focus on how exposure to others' reasoning influences preference and belief revision
- Built and maintained an MCP-based, two-phase deliberation workflow, supporting transcript extraction, reasoning summarization, and pilot-stage data archiving

Booth School of Business, The University of Chicago

Chicago, IL

Research Assistant, *Mindworks/the Perception and Judgment Lab*

July 2025 - Present

- Led backend system design for MyStyle, an interactive generative-AI exhibit that models individual preferences from real-time user ratings
- Architected and implemented Group Mode infrastructure (FastAPI + SQLAlchemy), enabling multi-user sessions, scoped data ingestion, preference aggregation, and backward-compatible integration with single-user workflows
- Built asynchronous modeling pipelines (Redis + RQ) and Dockerized deployment, supporting ingest/model/regenerate job queues, reproducible data isolation, and experiment-ready archival

Cognitive Science, The University of Chicago

Chicago, IL

Graduate Researcher, Advisor: Dr. Eugene Yu Ji (now at University of Waterloo & Mila Institute) *May 2025 - Present*

- Examined whether LLMs can function as socially competent participants in multi-party dialogue, focusing on interactional competence, indexicality, and register construction
- Designed a 2×3×2 factorial simulation framework and analyzed 1,200 GPT-4o responses using RoBERTa-based interactional strategy classification, SBERT embeddings, and UMAP-based register visualization
- Demonstrated that explicit role prompting constrains interactional strategy diversity, formal and informal registers separate in embedding space, and output-length constraints systematically reduce interactional richness

Business School, Beijing Normal University

Beijing, China

Undergraduate Dissertation

November 2023 - May 2024

- Evaluated the causal impact of a national policy initiative on industrial carbon emissions using synthetic control methods with panel data from 89 countries over 15 years

Project Facilitator, National Key Undergraduate Research Program

November 2022 - August 2023

- Led an experimental research project on carbon-footprint information interventions and food choice, examining how salience and social norms affect low-carbon consumption

- Designed the study, secured project approval, and coordinated participant recruitment and data collection across five experimental conditions ($N \approx 300$)
- Estimated treatment effects using regression analysis, finding that salient carbon information and social-norm cues significantly increased low-carbon food choices

Research Collaborator, Laboratory of Behavior and Decision

October 2021 – July 2025

Advisor: Prof. Haoran He

- Investigated how behavioral interventions shape clinicians' adoption of AI-assisted 3D reconstruction tools in medical decision-making
- Built and maintained interactive web-based experimental platforms to collect decision data

China Philanthropy Research Institute

Beijing, China

Research Assistant, Center for Philanthropic Studies

September 2022 - February 2024

- Constructed large-scale philanthropy datasets (1,000+ organizations, 100,000+ projects / records) for national reports on digital philanthropy in China
- Co-authored the *40th Anniversary of the Red Cross Society of China Shenzhen Branch* institutional research report

INVITED TALKS & PRESENTATIONS

Nudge Design and Low-Carbon Consumption

School of Public Policy and Management, UCAS; Business School, BNU; Antai College, SJTU

Spring 2023

HONORS, AWARDS & GRANTS

- \$5,000, University of Chicago, CAPP Startup Program, 2024 & 2025
- \$20,000, University of Minnesota (Twin Cities Campus), Minnesota Nonresident Tuition Awards (offered), 2024
- \$6,000, Ministry of Education of the PRC & Beijing Normal University, Undergraduate Research Grants, 2022
- Grand Prize, The 9th China National College Student Competition on Energy Economics (top 10 among 6148), 2023

TEACHING EXPERIENCE

Teaching Assistant

- CAPP 30254 Machine Learning for Public Policy, UChicago *Spring 2026*
- CAPP 30271 Mathematics for Computer Science and Data Analysis/Linear Algebra, UChicago *Winter 2026*
- Management Information Systems (MBA), Beijing Normal University *Spring 2024*

SELECTED PROJECTS

InterLines

November 2025 - Present

- Architected and implemented a multi-agent LLM system for iteratively structuring and refining research & policy documents into public-facing briefs
- Developed prompt orchestration, agent coordination, and modular pipelines for iterative refinement workflows and human-in-the-loop control

APPLIED RESEARCH & POLICY EXPERIENCE

World Institute of Politics, Economy and Technology (WPET)

Shanghai, China

Junior Research Fellow, Center for Technology Governance

October 2023 - Present

- Authored 30+ policy briefs and analytical reports on AI governance and digital regulation, translating comparative policy research into insights for public communication and internal policy workshops (*highest read count: 12,000+*)

LANGUAGES & SKILLS

- Programming & Statistical Analysis: Python (advanced); R and Stata for statistical modeling
- Interaction & Text Analysis: text-as-data methods; turn-level dialogue analysis; interactional strategy classification; embedding-based semantic analysis; clustering and dimensionality reduction
- AI Methods: LLM-based simulations; fine-tuning and evaluation of generative models; RLHF
- Systems & Data Infrastructure: backend development and data pipelines (APIs, relational/vector databases, Docker)
- Languages: English (Advanced), Chinese (Native), Spanish (Intermediate)